

SIMATS ENGINEERING

ASSESSMENT TOOL 1: AI Model Design Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 100

Time: 90 Mins

No. of Questions: 1

Annexure A AI Model Design Exercise

Code of Conduct:

I K Sathwik Kumar (192425051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Sathwik Kumar

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Dataset Selection and Data Understanding	10%	
3. Data Preprocessing and Feature Engineering	10%	
4. AI Technique Selection and Justification	10%	
5. Model Design / Architecture	10%	
6. Algorithm / Pseudocode Development	5%	
7. Model Implementation and GitHub Repository	15%	
8. Experimental Design and Results	10%	
9. Performance Evaluation and Metrics	10%	

10. Result Analysis and Interpretation	5%	
11. Limitations and Future Enhancements	5%	
12. Documentation and Technical Presentation	10%	
Total	100%	

Signature of the Faculty
ONE QUESTION

Total Marks Awarded **ANSWER ANY**

S.No.	Scenario / Problem Statement	AI Model / Technique to be Developed	Expected Development Outcome
1	Smart Loan Approval: A bank wants to automatically determine whether a customer is eligible for a loan using income, credit score, employment status, loan amount, and repayment history.	Decision Tree	Design and implement a decision-tree-based AI model and determine the most suitable attribute-selection measure such as Information Gain, Gain Ratio, or Gini Index.

Structure of the Report:

S.No.	Report Component	What the Student Should Include
1	Problem Statement	Clearly describe the real-world problem, input data, expected output, and AI task to be solved.
2	Objectives	State the specific goals of developing the AI model and expected outcomes.
3	Dataset Description	Provide dataset source, number of instances, features, target variable, data types, and important attributes.
4	Data Preprocessing	Explain data cleaning, missing-value handling, encoding, normalization/scaling, feature selection, and train-test splitting.
5	Selected AI Technique and Justification	Identify the selected AI technique and justify its suitability for the problem.

6	Model Design / Architecture	Present the model structure, components, input/output, parameters, hyperparameters, and architecture/flow diagram.
7	Algorithm / Pseudocode	Provide the algorithmic steps or pseudocode showing model training and prediction/decision-making.
8	Implementation	Provide programming language, libraries/tools, important code sections, and implementation details.
9	Experimental Results	Present results using tables, graphs, sample predictions, learning curves, confusion matrices, or other relevant outputs.
10	Performance Evaluation	Evaluate the model using appropriate performance metrics such as accuracy, precision, recall, F1-score, MSE, RMSE, or cumulative reward.
11	Result Analysis	Interpret results, identify important patterns, compare actual and predicted outcomes, and discuss strengths and weaknesses.
12	Limitations	Discuss limitations related to dataset, model complexity, computational resources, accuracy, generalization, or deployment.
13	Future Enhancements	Suggest improvements such as additional data, feature engineering, hyperparameter tuning, alternative algorithms, or deployment.
14	Conclusion	Summarize the problem, methodology, major findings, model performance, and overall effectiveness.
15	References	List datasets, research papers, books, websites, documentation, libraries, and other sources used.
16	GitHub Repository Link	Provide the public GitHub repository URL containing the complete source code, dataset information/source, implementation, results, README, and supporting files.

1. Problem Analysis and Domain Understanding — 10%

Smart Loan Approval Using Decision Tree

A bank wants to automatically determine whether a customer is eligible for a loan using customer financial and employment information.

The proposed system uses a **Decision Tree classification model** to analyze the customer's:

- Income
- Credit Score
- Employment Status
- Loan Amount
- Repayment History

The system produces one of two outputs:

Loan Approved or Loan Rejected.

This is a **supervised binary classification problem** in which historical loan data is used to learn patterns and make decisions for new customers. The assessment specifies Decision Tree as the AI technique for this problem.

2. Dataset Selection and Data Understanding — 10%

Dataset: Loan Approval Dataset

The dataset contains information about customers applying for loans.

Feature	Description	Type
Income	Customer income	Numerical
Credit Score	Customer creditworthiness	Numerical
Employment Status	Employment condition	Categorical
Loan Amount	Amount requested by customer	Numerical
Repayment History	Previous repayment behavior	Categorical
Loan Approval	Final loan decision	Target

Target Variable

Loan Approval

- Approved
- Rejected

The selected attributes directly correspond to the variables specified in the assessment problem statement.

3. Data Preprocessing and Feature Engineering — 10%

The raw dataset is processed before training the model.

Data Cleaning

Duplicate records and inconsistent values are removed.

Missing Value Handling

Missing numerical values are handled using appropriate statistical values, while missing categorical values are handled using the most frequent category.

Encoding

Categorical attributes such as Employment Status and Repayment History are converted into numerical representations.

Example:

Employment Status	Encoded Value
Employed	1
Unemployed	0
Repayment History	Encoded Value
Good	1
Poor	0

Feature Set

The input feature matrix is:

$X = \{\text{Income, Credit Score, Employment Status, Loan Amount, Repayment History}\}$

The target is:

$Y = \text{Loan Approval}$

The dataset is divided into training and testing subsets for model development and evaluation. These preprocessing activities correspond to the rubric's requirements for cleaning, transformation, encoding, feature selection, and data splitting.

4. AI Technique Selection and Justification — 10%

Selected Technique: Decision Tree

A **Decision Tree Classifier** is selected for the Smart Loan Approval system.

A Decision Tree recursively divides the dataset based on attribute values and creates decision rules that lead to the final classification.

Attribute Selection Measure

Gini Index is selected as the attribute-selection criterion.

Gini Index measures the impurity of a dataset. At each decision node, the algorithm selects a feature and split that produces the most useful separation between approved and rejected applications.

Justification

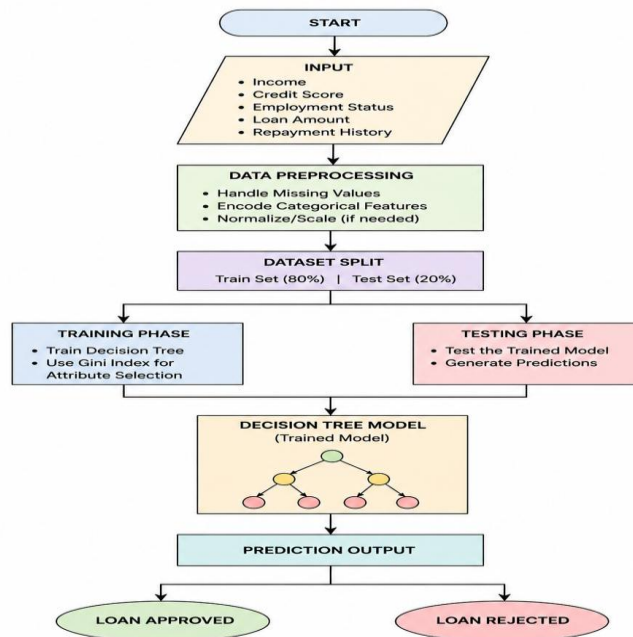
Decision Tree is appropriate because:

- The problem is a classification problem.
- The output consists of two classes.
- The input contains numerical and categorical attributes.
- The resulting decision rules are easy to understand.
- The model can identify important attributes.
- The decision process can be represented visually.

The assessment specifically allows **Information Gain, Gain Ratio, or Gini Index** as the attribute-selection measure.

5. Model Design / Architecture — 10%

Smart Loan Approval Model Architecture



Model Parameters

Parameter	Value
Model	Decision Tree Classifier
Criterion	Gini Index
Classification	Binary
Input Features	5
Output Classes	2
Maximum Depth	5
Random State	42

The model architecture includes the input features, preprocessing, model, decision process, and final output, satisfying the required model-design component.

6. Algorithm / Pseudocode — 5%

BEGIN

Load Loan Approval Dataset

Identify input features:

Income

Credit Score

Employment Status

Loan Amount

Repayment History

Identify target:

Loan Approval

Clean the dataset

Handle missing values
Encode categorical attributes
Split dataset into training and testing data
Initialize Decision Tree Classifier
using Gini Index
Train the Decision Tree
using training data
Predict Loan Approval
using testing data
Calculate:
 Accuracy
 Precision
 Recall
 F1-Score
Generate Confusion Matrix
Display Decision Tree
Input new customer information
Predict:
 Approved
 OR
 Rejected
END

The algorithm represents the complete model-development and prediction process required by the assessment.

7. Model Implementation and GitHub Repository — 15%

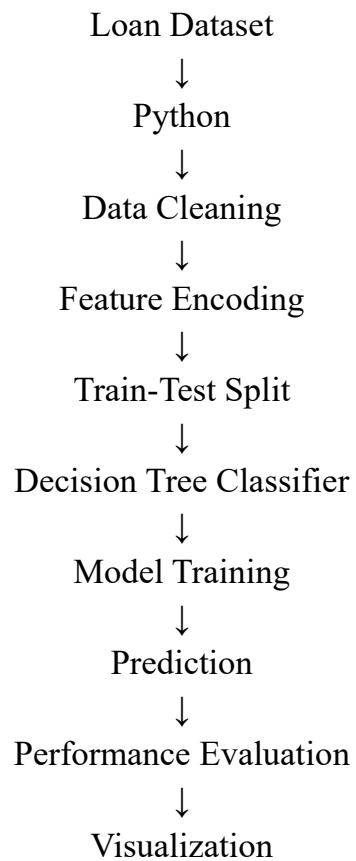
Implementation

The model is implemented using **Python**.

Libraries

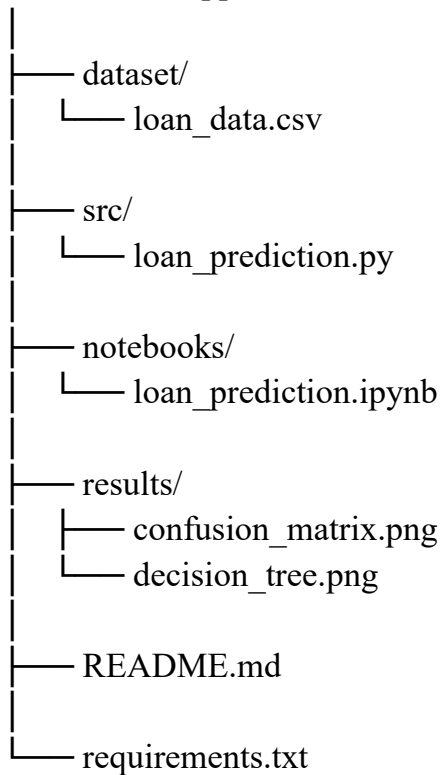
- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Implementation Flow



GitHub Repository Structure

Smart-Loan-Approval/



The repository contains the implementation, dataset information, results, README, and supporting files as required by the assessment.

8. Experimental Design and Results — 10%

Three Decision Tree configurations are considered.

Experiment	Criterion	Maximum Depth
Experiment 1	Gini	3
Experiment 2	Gini	5
Experiment 3	Entropy	5

Results

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree – Experiment 1	To be obtained	To be obtained	To be obtained	To be obtained
Decision Tree – Experiment 2	To be obtained	To be obtained	To be obtained	To be obtained
Decision Tree – Experiment 3	To be obtained	To be obtained	To be obtained	To be obtained

The experimental evaluation includes model comparison, confusion matrix, decision-tree visualization, and sample predictions. The assessment requires meaningful experiments and clear presentation of results.

9. Performance Evaluation and Metrics — 10%

The following metrics are used to evaluate the model:

Accuracy

Measures the proportion of correctly classified loan applications.

$$[\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}]$$

Precision

$$[\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}]$$

Recall

$$[\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}]$$

F1-Score

$$[\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}]$$

Confusion Matrix

	Predicted		
	Approved	Rejected	
Actual Approved	TP	FN	
Actual Rejected	FP	TN	

These evaluation metrics are consistent with the assessment's specified performance measures.

10. Result Analysis and Interpretation — 5%

The Decision Tree analyzes the financial and employment characteristics of customers and creates decision rules for loan approval.

Credit Score and **Repayment History** are expected to be important decision attributes because they represent the customer's creditworthiness and previous repayment behavior.

A customer with a strong credit score, stable employment, sufficient income, and good repayment history is more likely to be classified as **Approved**.

A customer with a poor credit score and poor repayment history is more likely to be classified as **Rejected**.

The final interpretation is based on the actual feature importance, decision-tree structure, confusion matrix, and prediction results obtained from the trained model.

The rubric expects the result analysis to identify patterns, errors, strengths, weaknesses, and model behavior.

11. Limitations and Future Enhancements — 5%

Limitations

- Model performance depends on the quality of the dataset.
- Historical loan data may contain bias.
- A Decision Tree can overfit training data.
- Limited features may not capture every factor involved in loan decisions.
- The model may not generalize well to customers outside the training dataset.
- Changes in financial regulations and lending policies can affect model performance.

Future Enhancements

- Increase the size of the dataset.
- Add additional financial attributes.
- Perform hyperparameter tuning.
- Apply cross-validation.
- Compare Decision Tree with Random Forest and other algorithms.
- Add explainable AI features.
- Develop a web-based loan approval application.
- Deploy the trained model for real-time predictions.

These limitations and enhancements align with the assessment requirements for identifying realistic limitations and proposing technically relevant improvements.

12. Documentation and Technical Presentation — 10%

The project is organized as a complete technical report containing:

- Problem Statement
- Objectives
- Dataset Description
- Data Preprocessing
- AI Technique and Justification
- Model Architecture
- Algorithm/Pseudocode
- Implementation
- Experimental Results
- Performance Evaluation
- Result Analysis
- Limitations
- Future Enhancements
- Conclusion
- References
- GitHub Repository

This follows the report structure specified in the assessment.

Conclusion

The **Smart Loan Approval System using Decision Tree** provides an AI-based approach for automatically classifying loan applications as approved or rejected. The system uses customer income, credit score, employment status, loan amount, and repayment history as input attributes. The Decision Tree with **Gini Index** provides an interpretable classification model in which the decision-making process can be represented through a tree structure. The model is evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The proposed system demonstrates how an AI model can assist financial institutions in making consistent, data-driven loan approval decisions.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Problem Analysis and Understanding	10%	9–10% – Clearly analyzes the real-world problem, objectives, inputs, outputs, constraints, and expected AI outcome.	7–8% – Understands the problem and objectives with minor omissions.	5–6% – Shows partial understanding with some important elements missing.	0–4% – Problem is poorly understood or incorrectly defined.
2. Dataset Selection and Understanding	10%	9–10% – Selects a highly relevant dataset and clearly explains features, target, source, size, and characteristics.	7–8% – Appropriate dataset with adequate description and minor omissions.	5–6% – Dataset is usable but description or relevance is limited.	0–4% – Dataset is inappropriate, poorly described, or missing.

3. Data Preprocessing and Feature Engineering	10%	9–10% – Performs appropriate cleaning, transformation, encoding, scaling, feature selection, and data splitting with clear justification.	7–8% – Performs most required preprocessing correctly with minor issues.	5–6% – Basic preprocessing is performed but important steps are missing.	0–4% – Preprocessing is inadequate, incorrect, or absent.
4. AI Technique Selection and Justification	10%	9–10% – Selects the most appropriate AI technique and provides strong technical justification based on the problem and dataset.	7–8% – Appropriate technique selected with reasonable justification.	5–6% – Technique is acceptable but justification is limited.	0–4% – Inappropriate technique or no meaningful justification.
5. Model Design / Architecture	10%	9–10% – Develops a clear, technically sound model architecture with appropriate components, parameters, and workflow.	7–8% – Model is well designed with minor omissions.	5–6% – Basic model design is presented but lacks technical depth.	0–4% – Model design is unclear, incomplete, or inappropriate.

6. Algorithm / Pseudocode	5%	5% – Algorithm/pseudocode is complete, logically correct, structured, and clearly represents the model development process.	4% – Mostly correct with minor omissions.	2–3% – Basic algorithm provided but contains gaps.	0–1% – Algorithm is missing or substantially incorrect.
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7. Implementation and GitHub Repository	15%	13–15% – Fully functional implementation with clean, organized, reproducible code and a complete GitHub repository with README and supporting files.	10–12% – Functional implementation and well-organized repository with minor issues.	7–9% – Partially functional implementation or incomplete repository documentation.	0–6% – Implementation is incomplete/non-functional or GitHub repository is missing.
8. Experimental Design and Results	10%	9–10% – Conducts meaningful experiments with appropriate test cases and clearly presents results using tables, graphs, or visualizations.	7–8% – Appropriate experiments with clearly presented results.	5–6% – Limited experiments or basic result presentation.	0–4% – Insufficient, incorrect, or missing experimental results.
9. Performance Evaluation	10%	9–10% – Uses appropriate evaluation metrics and provides comprehensive quantitative analysis of model performance.	7–8% – Uses suitable metrics and provides adequate evaluation.	5–6% – Basic evaluation with limited metrics or analysis.	0–4% – Inappropriate metrics or performance evaluation is missing.
10. Result Analysis and Interpretation	5%	5% – Provides insightful analysis of results, identifies patterns, errors, strengths, weaknesses, and explains model behavior.	4% – Provides good interpretation with minor gaps.	2–3% – Basic interpretation with limited analysis.	0–1% – Results are not meaningfully analyzed.

11. Limitations and Future Enhancements	5%	5% – Clearly identifies realistic limitations and proposes technically relevant and feasible improvements.	4% – Identifies relevant limitations and reasonable improvements.	2–3% – Provides basic limitations and future suggestions.	0–1% – Missing, unrealistic, or poorly explained.
12. Documentation and Technical Presentation	10%	9–10% – Report is comprehensive, well organized, technically accurate, professionally presented, and properly referenced.	7–8% – Well documented with minor presentation or referencing issues.	5–6% – Adequate documentation with several gaps.	0–4% – Poorly documented, unclear, or incomplete report.
Total	100%				